Root Cause Analysis (RCA) Generator System Documentation and Technical Reference Guide

🏆 Executive Summary

This internship focused on the development of AI-powered solutions under the AI Full Cycle Integration Project, particularly the RCA (Root Cause Analysis) Generator and Meeting Notes Summarizer. These systems were developed using FastAPI, LangGraph, LangChain, and Gemini AI to automate report generation, meeting documentation, and workflow analysis, helping improve efficiency, consistency, and accuracy in business processes.

In addition to system development, extensive testing, system exploration, and defect documentation were conducted on the Navee HRIS platform. Various modules were evaluated to identify issues, validate functionality, and support quality assurance efforts. The project provided valuable experience in artificial intelligence integration, software development, API implementation, testing, and technical documentation within a real-world business environment.

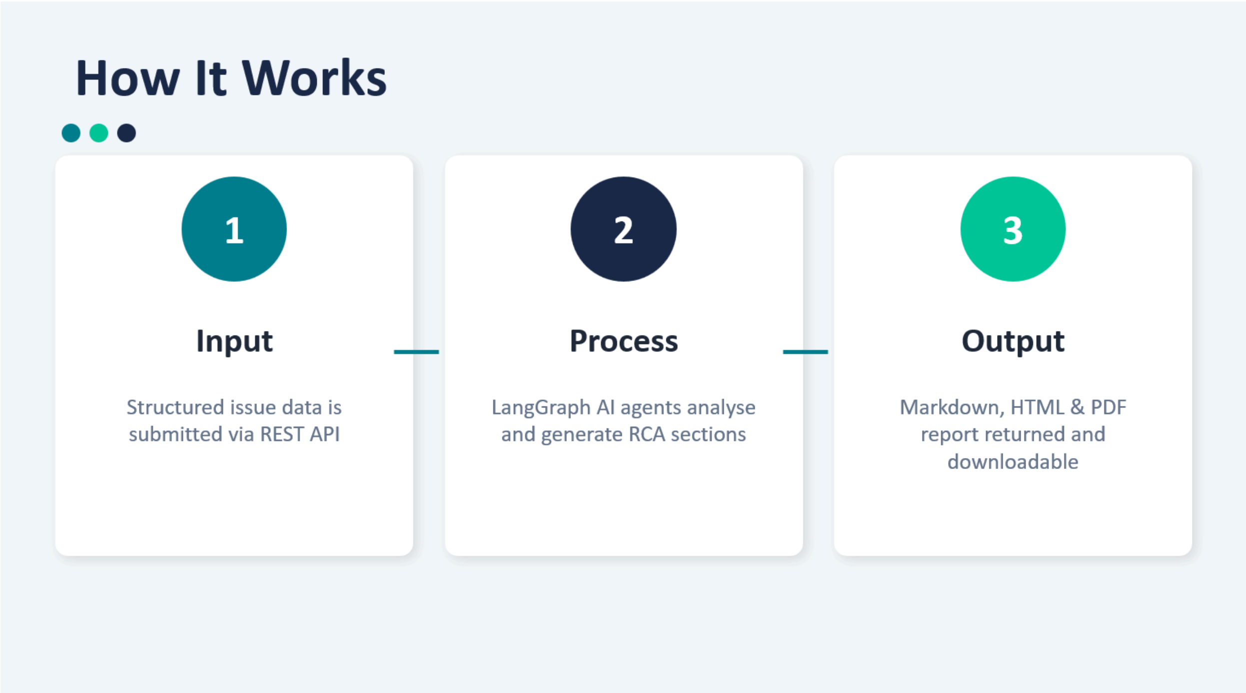
📌 Objectives

To develop and evaluate AI-powered solutions that automate business processes, improve documentation accuracy, and enhance operational efficiency through the integration of modern software development and artificial intelligence technologies.

🎁 Specific Objectives

- To develop an RCA (Root Cause Analysis) Generator that automates the creation of structured incident analysis reports.
- To develop a Meeting Notes Summarizer that converts meeting recordings into organized Minutes of Meeting (MOM) documents.
- To implement secure and scalable APIs using FastAPI, LangGraph, and related technologies.
- To conduct system exploration, testing, and defect identification on the Navee HRIS platform.
- To apply software quality assurance practices through validation, testing, and documentation.
- To gain practical experience in AI integration, workflow automation, and real-world software development processes.

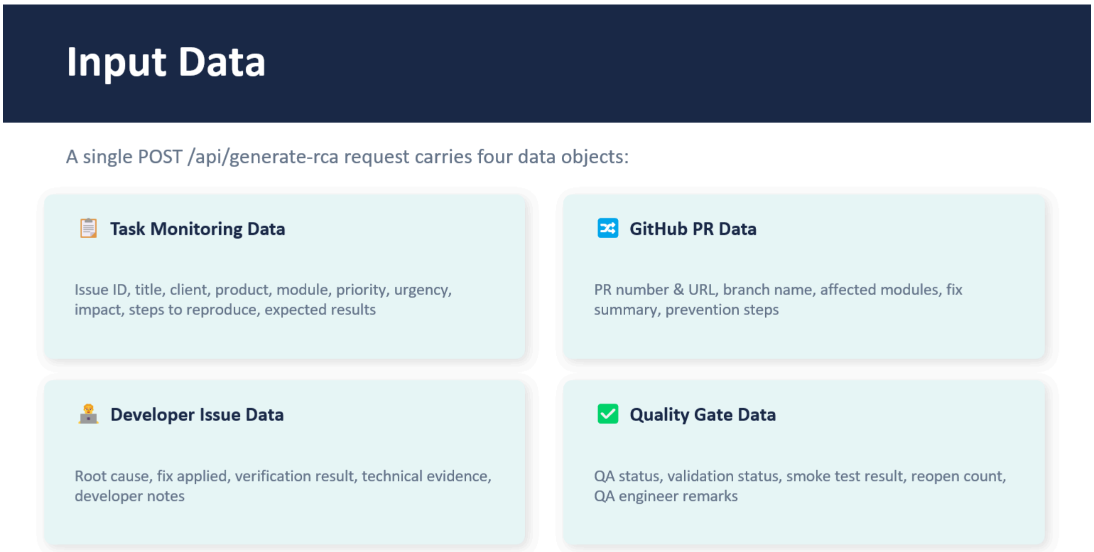
📍 System Architecture



The system architecture follows an AI-powered backend workflow using **FastAPI** as the main API framework. User input is submitted through **Swagger UI**, validated using **Pydantic models**, and processed by the backend service layer. The system uses **LangChain** and **LangGraph** to manage AI workflows, connect agents, and control the step-by-step generation process.

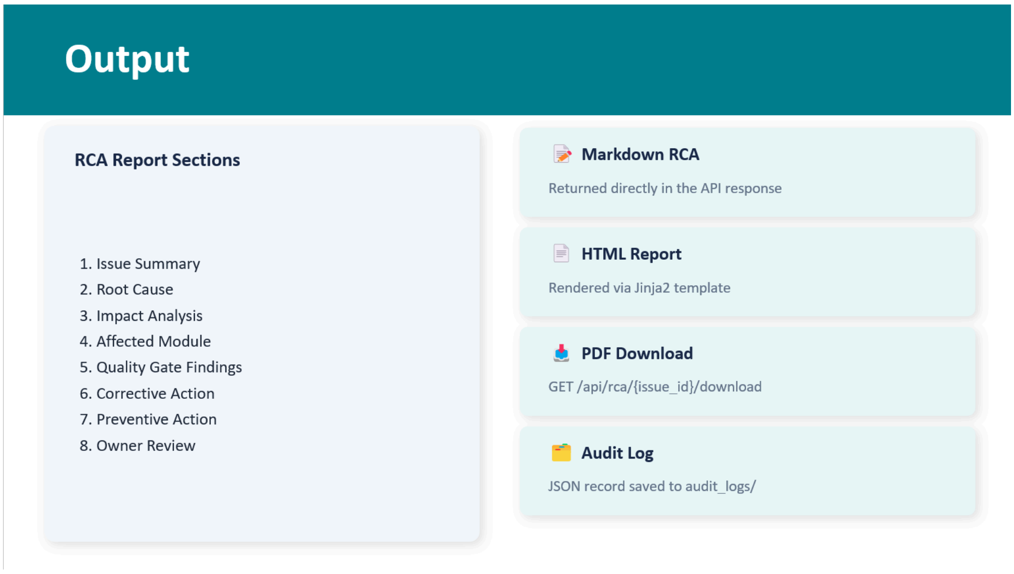
For the RCA Generator and Meeting Notes Summarizer, the system sends structured data or transcripts to **Gemini AI** for content generation. The generated output is reviewed, formatted into **HTML**, and converted into **PDF** for final documentation. Additional components such as **API key authentication**, **rate limiting**, **audit logging**, and **LangSmith tracing** are included to improve security, monitoring, and reliability.

🌰 Workflow



The system workflow starts when the user submits data through the API or Swagger UI. The input is validated using Pydantic, then passed to the service layer where LangGraph controls the step-by-step AI process.

For the RCA Generator, the workflow analyzes issue details, technical impact, QA results, and prevention actions before generating the final RCA report.



⛱️ Features

The RCA Generator is an AI-powered system designed to automate the creation of Root Cause Analysis (RCA) reports for recurring client issues and system defects. The system provides the following features:

- Automated Root Cause Analysis report generation
- Multi-agent AI workflow orchestration using LangGraph
- Incident analysis and issue assessment
- Technical impact evaluation
- Quality Gate and QA validation review
- Corrective and preventive action recommendations
- Owner review and resolution documentation
- Structured input validation using Pydantic
- AI-powered content generation using Gemini AI
- API key authentication and authorization
- Rate limiting and security controls
- Audit logging and activity tracking
- Health monitoring and provider status checks
- HTML report generation
- Professional PDF report generation

- Swagger UI integration for testing and API documentation
- Fallback and error-handling mechanisms for AI service interruptions
- End-to-end workflow automation from input validation to final report generation

🥇 Limitations

While the RCA Generator significantly improves the speed and consistency of Root Cause Analysis creation, several limitations remain that should be considered during its use and future development.

📍 Key Limitations

- Dependent on AI provider availability and API quota limits.
- Requires accurate and complete input data to generate reliable results.
- Generated RCA reports may still require human review and approval.
- GitHub PR integration is currently limited and not fully automated.
- No database implementation for long-term storage and report management.
- AI-generated content may occasionally require refinement for complex technical issues.
- PDF generation depends on proper HTML, CSS, and template configuration.

2 ROOT CAUSE

Generation unavailable due to temporary AI provider limitations. Please retry the request or contact support if this persists.

3 AFFECTED MODULE

Generation unavailable due to temporary AI provider limitations. Please retry the request or contact support if this persists.

4 IMPACT ANALYSIS

Generation unavailable due to temporary AI provider limitations. Please retry the request or contact support if this persists.

🎼 Current Scope Constraints

The system is currently designed primarily for internal use, testing, and demonstration purposes. Although security features such as API authentication and rate limiting have been implemented, additional production-level enhancements, monitoring, and scalability improvements may be required before enterprise deployment.

🏝️ AI-Related Constraints

As the system relies on Large Language Models (LLMs), output quality may vary depending on the context provided. During periods of high AI service demand, fallback mechanisms can maintain functionality; however, the generated content may be less detailed compared to the primary AI model.

🥛 Future Enhancements

The RCA Generator can be further enhanced to improve its accuracy, scalability, and usability. Future development efforts may focus on expanding integrations, strengthening automation, and improving report quality.

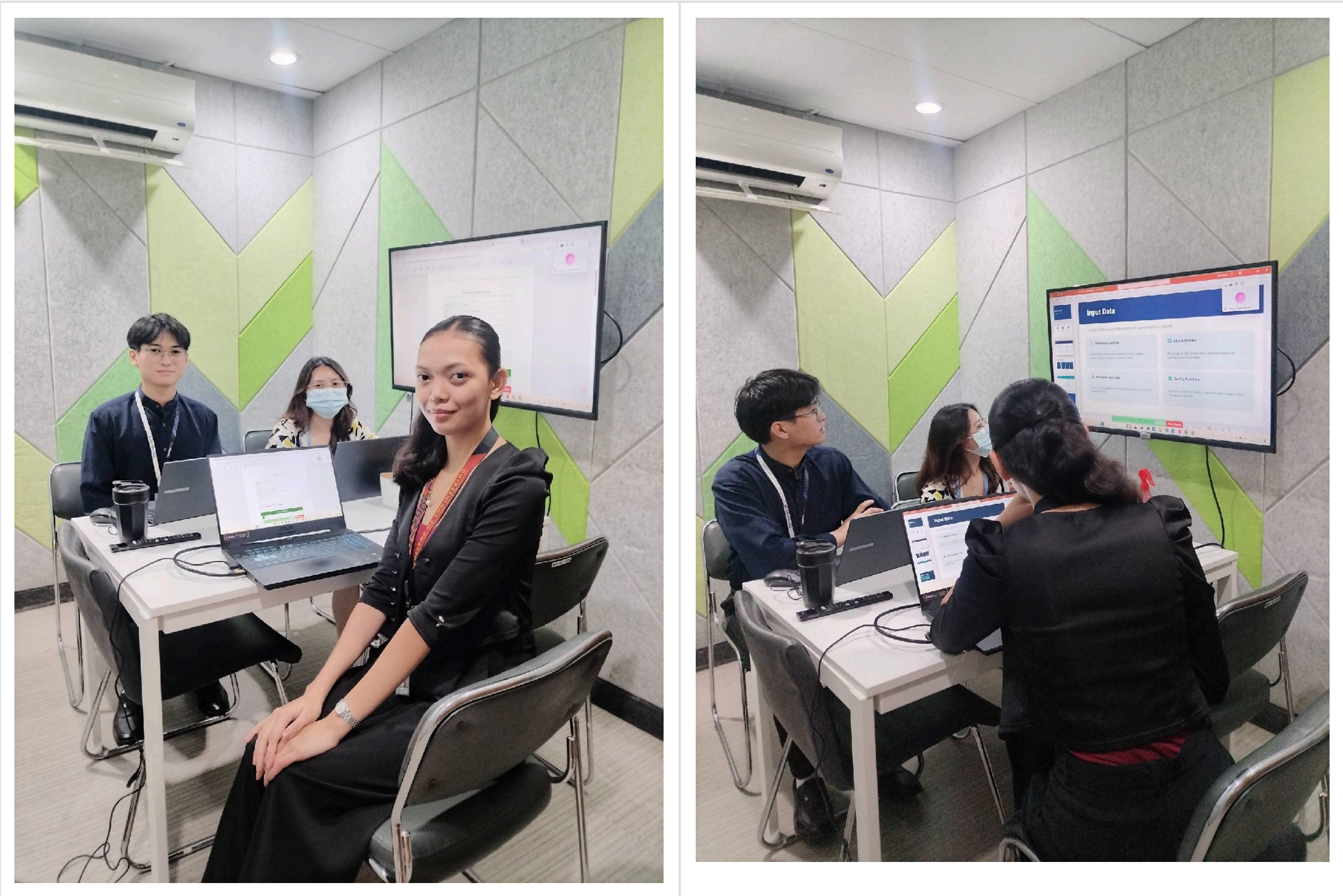
🥇 Proposed Enhancements

- Integration with GitHub Pull Requests for automatic issue and code analysis.
- Database implementation for RCA history, tracking, and report management.
- Multi-provider AI support for improved reliability and failover capabilities.
- Advanced analytics and dashboards for RCA trends and recurring issue monitoring.
- Automated email notifications and report distribution.
- Enhanced PDF templates with customizable company branding.
- Role-based access control and user management.
- Integration with ticketing and project management systems.
- Improved AI prompting and validation mechanisms for higher report accuracy.
- Deployment to cloud infrastructure for better scalability and availability.

🎨 Long-Term Vision

Future versions of the system may evolve into a centralized incident management and quality assurance platform capable of automatically collecting issue data, generating RCA reports, tracking resolutions, and providing actionable insights to support continuous process improvement and decision-making.

🎉 DOCUMENTATION



Presentation Day

This presentation provides an overview of the RCA (Root Cause Analysis) Generator, an AI-powered system developed to automate the creation of RCA reports for recurring client issues and system defects. The discussion covers the project's objectives, system architecture, workflow, key features, and technical implementation using FastAPI, LangGraph, LangChain, and Gemini AI.

A live system walkthrough is also included to demonstrate the complete end-to-end process, from submitting input data through Swagger UI, processing information through the AI workflow, generating RCA content, and producing the final PDF report. Additionally, the presentation highlights the code structure, major components, testing activities, current limitations, and future enhancement opportunities for the system.

📚 GENERATED OUTPUT

4F. Victoria One Bldg. 1670 Quezon Ave.
South Triangle, Quezon City, Philippines.
www.direcbusiness.com
(02) 8555 1547

ROOT CAUSE ANALYSIS (RCA)

Company Name:	**Direc Business Technologies Inc.**
Purpose:	Root Cause Analysis (RCA)
Date:	June 18, 2026
Issue ID:	EIL_TEST_001
Issue:	Attendance records not updating after biometric synchronization
Assigned Consultant:	Jane Arianne A. Gaela
Approval Status:	*For Review*

ISSUE NO. 1

Attendance records not updating after biometric synchronization

PRODUCT People Navee HRIS | MODULE Attendance Management | CORE FUNCTION Biometric Synchronization Service

ROOT CAUSE ANALYSIS

1 ISSUE

The Human Resource department reported an issue where employee attendance records were not being updated after the scheduled biometric synchronization process. Employees were able to successfully clock in and clock out using the biometric devices. However, the attendance information was not properly displayed inside the HRIS Attendance Management module. The issue caused inconsistencies between the physical biometric logs and the records stored in the system database.

PRODUCT	MODULE	CORE FUNCTION
People Navee HRIS	Attendance Management	Biometric Synchronization Service

2 ROOT CAUSE

TECHNICAL ROOT CAUSE ANALYSIS

After conducting an investigation, the development team identified that the failure occurred within the backend synchronization process responsible for transferring biometric device logs into the HRIS database.

The synchronization service encountered timeout errors whenever a large volume of employee attendance records was processed. The API connection was automatically terminated before the system completed the transaction.

Several contributing factors were discovered:

- The biometric API timeout configuration was too short for large attendance batches.
- The system processed records sequentially instead of using optimized batch processing.
- There was no automatic retry mechanism when synchronization failed.
- Error monitoring did not immediately notify administrators about failed synchronization attempts.

The issue was classified as a backend service reliability problem affecting data synchronization.

3 AFFECTED MODULE

AFFECTED SYSTEM COMPONENTS

The issue affected multiple HRIS components that depend on accurate attendance information.

- Attendance Monitoring Dashboard
- Employee Daily Time Records
- Biometric Integration Service
- Payroll Attendance Validation
- Attendance Report Generation

Other HRIS modules remained operational and no employee profile information was affected.

4 IMPACT ANALYSIS

BUSINESS IMPACT ANALYSIS

The synchronization failure affected daily HR operations because attendance records required manual verification before processing.

Employees who successfully recorded their attendance through the biometric device appeared as absent or incomplete in the system.

Potential operational impacts included:

- Incorrect attendance monitoring results
- Possible payroll computation discrepancies
- Additional manual checking workload for HR personnel
- Delay in attendance approval workflow
- Reduced reliability of automated reporting

5 QUALITY GATE FINDINGS

QUALITY GATE — FIRST PASS	SMOKE TEST — FIRST PASS	REOPEN COUNT
Failed	**Passed**	**1**
FC FAILED TESTING	**QA STATUS**	**QA VALIDATED BY**
Yes	**Validated After Fix**	**Jane Arianne A. Gaela**

6 CORRECTIVE ACTION

CORRECTIVE ACTION IMPLEMENTED

The development team modified the synchronization service to improve performance, reliability, and failure handling.

Implemented solutions:

- Increased biometric API timeout limit from 30 seconds to 120 seconds.
- Implemented retry mechanism using exponential backoff strategy.
- Optimized attendance processing using database batch transactions.
- Added detailed error logging for failed synchronization attempts.
- Created validation checks before marking synchronization as complete.

After deployment, the synchronization process was tested with large attendance datasets and completed successfully.

7 PREVENTIVE ACTION

PREVENTIVE ACTION PLAN

Preventive improvements were planned to avoid similar issues in future releases.

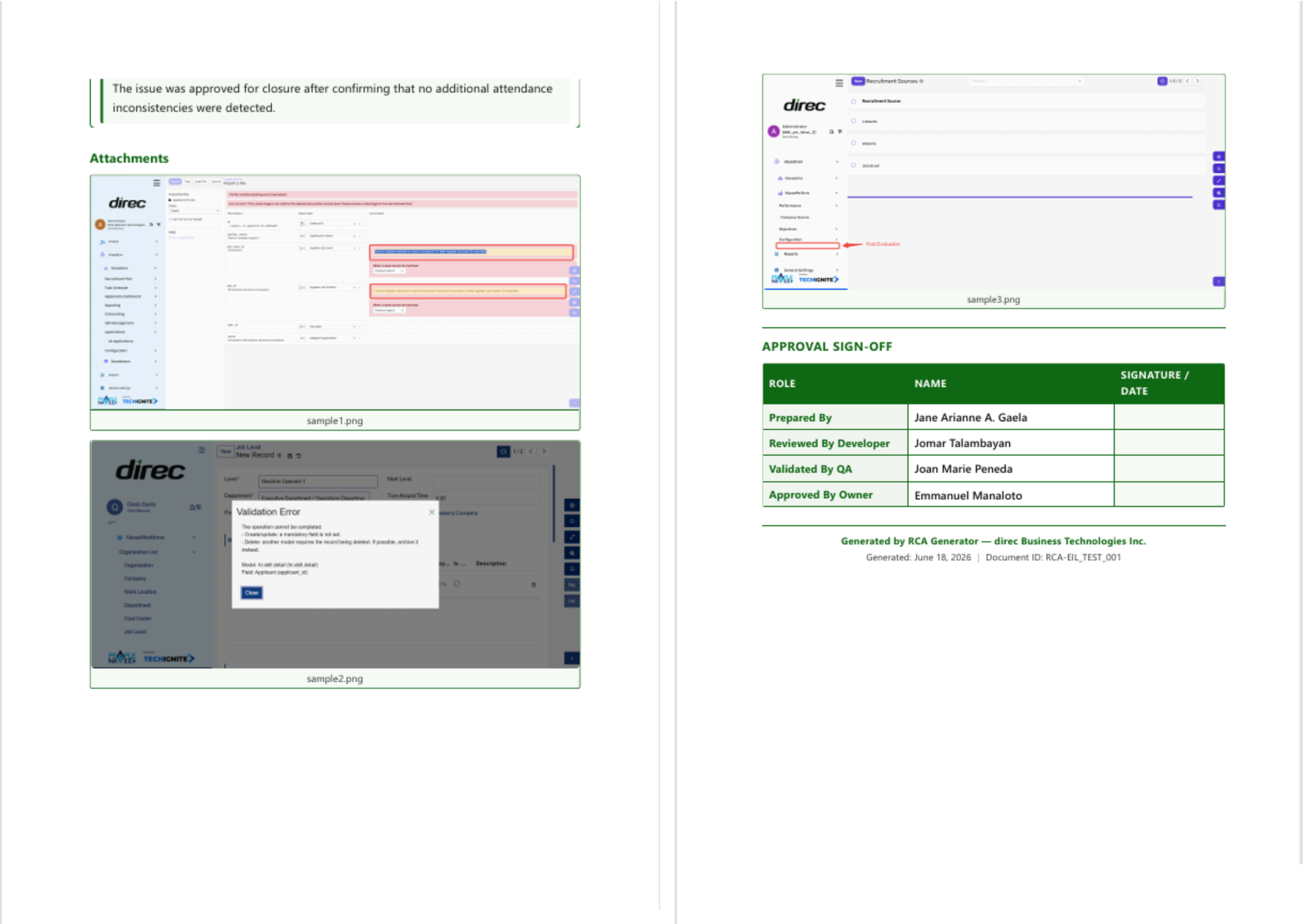
- Implement automated synchronization monitoring alerts.
- Create daily attendance reconciliation reports.
- Add automated regression test cases for biometric integration.
- Perform scheduled performance testing on high-volume transactions.
- Improve API failure handling documentation.

8 OWNER REVIEW

SYSTEM OWNER REVIEW

The system owner reviewed the implemented corrective actions, QA validation results, and testing evidence.

The solution successfully restored attendance synchronization functionality and improved system reliability.



> The issue was approved for closure after confirming that no additional attendance inconsistencies were detected.

Attachments

sample1.png

sample2.png

sample3.png

APPROVAL SIGN-OFF

ROLE	NAME	SIGNATURE / DATE
Prepared By	Jane Arianne A. Gaela	
Reviewed By Developer	Jomar Talambayan	
Validated By QA	Joan Marie Peneda	
Approved By Owner	Emmanuel Manaloto	

Generated by RCA Generator — direc Business Technologies Inc.
Generated: June 18, 2026 | Document ID: RCA-EIL_TEST_001

🎹 CONCLUSION

The development of the RCA Generator successfully demonstrated the application of artificial intelligence and modern software development practices in automating Root Cause Analysis reporting. By integrating FastAPI, LangGraph, LangChain, and Gemini AI, the system is capable of generating structured, professional RCA reports while reducing manual effort and improving documentation consistency.

Through the implementation, testing, and presentation of the system, valuable experience was gained in AI integration, API development, workflow automation, security, and quality assurance. While there are opportunities for future enhancements, the project has achieved its primary objectives and provides a strong foundation for further development and potential organizational adoption.